10. Cosets and the Theorem of Lagrange

1. (See the answer in the text.)

2. $4\mathbb{Z} = \{\cdots, -8, -4, 0, 4, 8, \cdots\}$, $2 + 4\mathbb{Z} = \{\cdots, -6, -2, 2, 6, 10, \cdots\}$

3. $\langle 2\rangle = \{0, 2, 4, 6, 8, 10\}$, $1 + \langle 2\rangle = \{1, 3, 5, 7, 9, 11\}$

4. $\langle 4\rangle = \{0, 4, 8\}$, $1 + \langle 4\rangle = \{1, 5, 9\}$, $2 + \langle 4\rangle = \{2, 6, 10\}$, $3 + \langle 4\rangle = \{3, 7, 11\}$

5. $\langle 18\rangle = \{0, 18\}$, $1 + \langle 18\rangle = \{1, 19\}$, $2 + \langle 18\rangle = \{2, 20\}$, $\cdots$, $17 + \langle 18\rangle = \{17, 35\}$

6. $\{\rho_0, \mu_2\}$, $\{\rho_1, \delta_2\}$, $\{\rho_2, \mu_1\}$, $\{\rho_3, \delta_1\}$

7. $\{\rho_0, \mu_2\}$, $\{\rho_1, \delta_1\}$, $\{\rho_2, \mu_1\}$, $\{\rho_3, \delta_2\}$ They are not the same.

8. We do not get a coset group. The 2×2 blocks in the table do not all have elements of just one coset.

	ρ_0	μ_2	ρ_1	δ_2	ρ_2	μ_1	ρ_3	δ_1
ρ_0	ρ_0	μ_2	ρ_1	δ_2	ρ_2	μ_1	ρ_3	δ_1
μ_2	μ_2	ρ_0	δ_1	ρ_3	μ_1	ρ_2	δ_2	ρ_1
ρ_1	ρ_1	δ_2	ρ_2	μ_1	ρ_3	δ_1	ρ_0	μ_2
δ_2	δ_2	ρ_1	μ_2	ρ_0	δ_1	ρ_3	μ_1	ρ_2
ρ_2	ρ_2	μ_1	ρ_3	δ_1	ρ_0	μ_2	ρ_1	δ_2
μ_1	μ_1	ρ_2	δ_2	ρ_1	μ_2	ρ_0	δ_1	ρ_3
ρ_3	ρ_3	δ_1	ρ_0	μ_2	ρ_1	δ_2	ρ_2	μ_1
δ_1	δ_1	ρ_3	μ_1	ρ_2	δ_2	ρ_1	μ_2	ρ_0

9. $\{\rho_0, \rho_2, \}$, $\{\rho_1, \rho_2\}$, $\{\mu_1, \mu_2\}$, $\{\delta_1, \delta_2, \}$

10. The same cosets are obtained as in Exercise 9, so the right cosets of $\{\rho_0, \rho_2\}$ are the same as the left cosets.

11. (See the answer in the text.)

12. $\langle 3\rangle = \{1, 3, 6, 9, 12, 15, 18, 21\}$ has 8 elements, so its index (the number of cosets) is $24/8 = 3$.

13. $\langle \mu_1\rangle = \{\rho_0, \mu_1\}$ has 2 elements, so its index (the number of left cosets) is $6/2 = 3$.

14. $\langle \mu_2\rangle = \{\rho_0, \mu_2\}$ has 2 elements, so its index (the number of left cosets) is $8/2 = 4$.

15. $\sigma = (1, 2, 5, 4)(2, 3) = (1, 2, 3, 5, 4)$ generates a cyclic subgroup of S_5 of order 5, so its index (the number of left cosets) is $5!/5 = 4! = 24$.

16. $\mu = (1, 2, 4, 5)(3, 6)$ generates a cyclic subgroup of S_6 of order 4, (the cycles are disjoint) so its index (the number of left cosets) is $6!/4 = 720/4 = 180$.

17. The definition is incorrect; we have no concept of a left coset of an arbitrary subset of a group G.

Let G be a group and let $H \leq G$. The **left coset of** H **containing** a is $aH = \{ah \mid h \in H\}$.

18. The definition is correct.

19. T T T F T F T T F T (i) See the last sentence in this section.

20. This is impossible. For a subgroup H of an abelian group G, we have $a + H = H + a$ for all $a \in G$.

21. For any group G, just take the subgroup $H = G$. **22.** The subgroup $\{0\}$ of $\mathbb{Z}_6$.

23. This is impossible. Because the cells are disjoint and nonempty, their number cannot exceed the order of the group.

24. This is impossible. The number of cells must divide the order of the group, and 4 does not divide 6.

25. The left cosets of the subgroup H form a partition of G and each coset has the same number of elements as H has.

26. *Reflexive:* Let $a \in G$. then $aa^{-1} = e$ and $e \in H$ because H is a subgroup. Thus $a \sim_R a$.

Symmetric: Suppose $a \sim_R b$. Then $ab^{-1} \in H$. Because H is a subgroup, $(ab^{-1})^{-1} = ba^{-1}$ is in H, so $b \sim_R a$.

Transitive: Suppose $a \sim_R b$ and $b \sim_R c$. Then $ab^{-1} \in H$ and $bc^{-1} \in H$. Because H is a subgroup $(ab^{-1})(bc^{-1}) = ac^{-1}$ is in H, so $a \sim_R c$.

27. Let $\phi_g : H \to Hg$ by $\phi_g(h) = hg$ for all $h \in H$. If $\phi_g(h_1) = \phi_g(h_2)$ for $h_1, h_2 \in H$, then $h_1 g = h_2 g$ and $h_1 = h_2$ by group cancellation, so ϕ_g is one to one. Clearly ϕ_g is onto Hg, for if $hg \in Hg$, then $\phi_g(h) = hg$.

28. We show that $gH = Hg$ by showing that each coset is a subset of the other. Let $gh \in gH$ where $g \in G$ and $h \in H$. Then $gh = ghg^{-1}g = [(g^{-1})^{-1}hg^{-1}]g$ is in Hg because $(g^{-1})^{-1}hg^{-1}$ is in H by hypothesis. Thus gH is a subset of Hg.

Now let $hg \in Hg$ where $g \in G$ and $h \in H$. Then $hg = gg^{-1}hg = g(g^{-1}hg)$ is in gH because $g^{-1}hg$ is in H by hypothesis. Thus Hg is a subset of gH also, so $gH = Hg$.

29. Let $h \in H$ and $g \in G$. By hypothesis, $Hg = gH$. Thus $hg = gh_1$ for some $h_1 \in H$. Then $g^{-1}hg = h_1$, showing that $g^{-1}hg \in H$.

30. It is false. Let $G = S_3, H = \{\rho_0, \mu_1\}, a = \rho_1$ and $b = \mu_3$. (See Table 8.8.) Then $aH = \{\rho_1, \mu_3\} = bH$, but $Ha = \{\rho_1, \mu_2\}$ while $Hb = \{\rho_2, \mu_3\}$.

31. It is true; $b = eb$ and $e \in H$ so $b \in Hb$. Because $Hb = Ha$, we have $b \in Ha$.

32. It is true. Because H is a subgroup, we have $\{h^{-1} \mid h \in H\} = H$. Therefore $Ha^{-1} = \{ha^{-1} \mid h \in H\} = \{h^{-1}a^{-1} \mid h \in H\} = \{(ah)^{-1} \mid h \in H\}$. That is, Ha^{-1} consists of all inverses of elements in aH. Similarly, Hb^{-1} consists of all inverses of elements in bH. Because $aH = bH$, we must have $Ha^{-1} = Hb^{-1}$.

33. It is False. Let H be the subgroup $\{\rho_0, \mu_2\}$ of D_4 in Table 8.12. Then $\rho_1 H = \delta_2 H = \{\rho_1, \delta_2\}$, and $\rho_1^2 H = \rho_2 H = \{\rho_2, \mu_1\}$ but $\delta_2^2 H = \rho_0 H = H = \{\rho_0, \mu_2\}$.

34. The possible orders for a proper subgroup are p, q, and 1. Now p and q are primes and every group of prime order is cyclic, and of course every group of order 1 is cyclic. Thus every proper subgroup of a group of order pq must be cyclic.

35. From the proof in Exercise 32, $Ha^{-1} = \{(ah)^{-1} \mid h \in H\}$ This shows that the map ϕ of the collection of left cosets into the collection of right cosets defined by $\phi(aH) = Ha^{-1}$ is well defined, for if $aH = bH$, then $\{(ah)^{-1} \mid h \in H\} = \{(bh)^{-1} \mid h \in H\}$. Because Ha^{-1} may be any right coset of H, the map is onto the collection of right cosets. Because elements in disjoint sets have disjoint inverses, we see that ϕ is one to one.

36. Let G be abelian of order $2n$ where n is odd. Suppose that G contains two elements, a and b, of order 2. Then $(ab)^2 = abab = aabb = ee = e$ and $ab \neq e$ because the inverse of a is a itself. Thus ab also has order 2. It is easily checked that then $\{e, a, b, ab\}$ is a subgroup of G of order 4. But this is impossible because n is odd and 4 does not divide $2n$. Thus there can't be two elements of order 2.

37. Let G be of order ≥ 2 but with no proper nontrivial subgroups. Let $a \in G, a \neq e$. Then $\langle a \rangle$ is a nontrivial subgroup of G, and thus must be G itself. Because every cyclic group not of prime order has proper subgroups, we see that G must be finite of prime order.

38. Following the hint and using the notation there, it suffices to prove $\{(a_i b_j)K \mid i = 1, \cdots, r, j = 1, \cdots, s\}$ is the collection of distinct left cosets of K in G. Let $g \in G$ and let g be in the left coset $a_i H$ of H. Then $g = a_i h$ for some $h \in H$. Let h be in the left coset $b_j K$ of K in H. Then $h = b_j k$ for some $k \in K$, so $g = a_i b_j k$ and $g \in a_i b_j K$. This shows that the collection given in the hint includes all left cosets of K in G. It remains to show the cosets in the collection are distinct. Suppose that $a_i b_j K = a_p b_q K$, so that $a_i b_j k_1 = a_p b_q k_2$ for some $k_1, k_2 \in K$. Now $b_j k_1 \in H$ and $b_q k_2 \in H$. Thus a_i and a_p are in the same left coset of H, and therefore $i = p$ and $a_i = a_p$. Using group cancellation, we deduce that $b_j k_1 = b_q k_2$. But this means that b_j and b_q are in the same left coset of K, so $j = q$.

39. The partition of G into left cosets of H must be H and $G - H = \{g \in G \mid g \notin H\}$, because G has finite order and H must have half as many elements as G. For the same reason, this must be the partition into right cosets of H. Thus every left coset is also a right coset.

40. Let $a \in G$. Then $\langle a \rangle$ has order d that must divide the order of G, so that $n = dq$. We know that a^d = e. Thus $a^n = (a^d)^q = e^q = e$ also.

41. Let $r + \mathbb{Z}$ be a left coset of $\mathbb{Z}$ in $\mathbb{R}$, where $r \in \mathbb{R}$. Let $[r]$ be the greatest integer less than or equal to r. Then $0 \leq r - [r] < 1$ and $r + (-[r])$ is in $r + \mathbb{Z}$. Because the difference of any two distinct elements in $r + \mathbb{Z}$ is at least 1, we see that $x = r - [r]$ must be the unique element $x \in r + \mathbb{Z}$ satisfying $0 \leq x < 1$.

42. Consider a left coset $r + \langle 2\pi \rangle$ of $\langle 2\pi \rangle$ in $\mathbb{R}$. Then every element of this coset is of the form $r + n(2\pi)$ for $n \in \mathbb{Z}$. We know that $\sin(r + n(2\pi)) = \sin r$ for all $n \in \mathbb{Z}$ because the function *sine* is periodic with period 2π. Thus *sine* has the same value at each elements of the coset $r + \langle 2\pi \rangle$.

43. **a.** *Reflexive:* We have $a = eae$ where $e \in H$ and $e \in K$, so $a \sim a$.

Symmetric: Let $a \sim b$ so $a = hbk$ for some $h \in H, k \in K$. Then $b = h^{-1} a k^{-1}$ and $h^{-1} \in H$ and $k^{-1} \in K$ because H and K are subgroups Thus $b \sim a$.

Transitive: Let $a \sim b$ and $b \sim c$ so $a = hbk$ and $b = h_1 c k_1$ for some $h, h_1 \in H$ and $k, k_1 \in K$. Then $a = hh_1 ck_1 k$ and $hh_1 \in H$ and $k_1 k \in K$ because H and K are subgroups. Thus $a \sim c$.

b. The equivalence class containing the element a is $HaK = \{hak \mid h \in H, k \in K\}$. It can be formed by taking the union of all right cosets of H that contain elements in the left coset aK.

44. **a.** *Closure:* If $\sigma(c) = c$ and $\mu(c) = c$, then $(\sigma\mu)(c) = \sigma(\mu(c)) = \sigma(c) = c$, so $S_{c,\,c}$ is closed under permutation multiplication.

Identity: The identity permutation leaves c fixed so it is in $S_{c,\,c}$.

Inverses: If σ leaves c fixed, then σ^{-1} does also. Thus $S_{c,\,c}$ is a subgroup of S_A.

b. No, $S_{c,\,d}$ is not closed under permutation multiplication. If $\sigma, \mu \in S_{c,\,d}$, then $(\sigma\mu)(c) = \sigma(\mu(c)) = \sigma(d)$. Because $\sigma(c) = d$ and σ is one to one, we know that $\sigma(d) \neq d$ unless $c = d$.